# EGX360 - Intelligent Stock Market Analysis and Forecasting Platform

AI-Powered EGX30 Forecasting Platform

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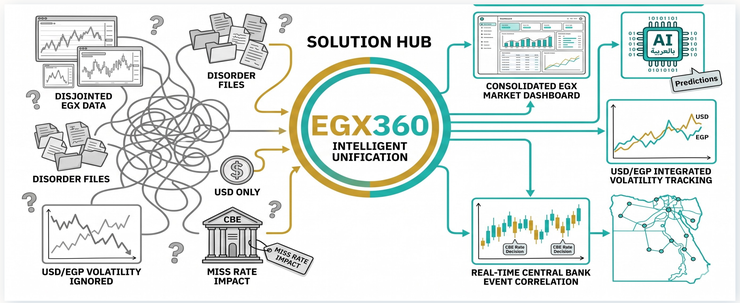
## Aims & Rationale

**The Problem**: Fragmented EGX data lacks unified, real-time AI analysis.

**The Goal**: EGX360: High-accuracy forecasting using a Stacking Ensemble

**Rationale**: Cloud-native architecture ensures scalable, high-performance processing.




## Summary of Work

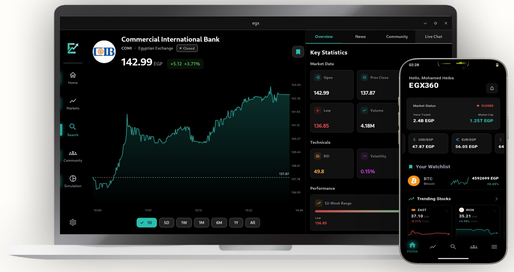
**Cross-Platform** UI Adaptive Flutter dashboard for Mobile and Desktop.

**AI Core** Stacking Ensemble (XGBoost + LightGBM) for trend forecasting.

**Data Infrastructure** Automated AWS pipeline for real-time data ingestion.

**Backend Services** Supabase integration for secure, high-performance.

**NLP Pipeline** Real-time sentiment extraction via Hugging Face.

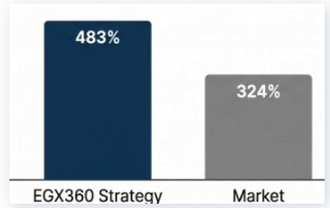


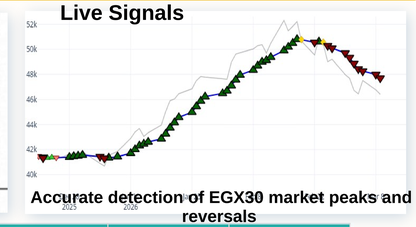

aws Flutter python

## Results & Performance

**+159.09%** Backtest Alpha

**90%** Model Accuracy




Accurate detection of EGX30 market peaks and reversals

| Study | Journal / Rank | Model Architecture | Market Context | Accuracy |
|---|---|---|---|---|
| Livieris et al. (2025) | Springer (Q1) | SVM, LSTM, GB | 55 Global Markets | 51% |
| Park et al. (2024) | MDPI (Q2) | LSTM vs GRU | NASDAQ | 65% |
| Chaudhari et al. (2025) | ScienceDirect (Q2) | Ensemble + Sentiment | BSE/NSE India | 72% |
| EGX360 (This Work) | KSIU Senior Project | Stacking (XGB + LGBM) | EGX30 (Egypt) | 90% |

## System Architecture

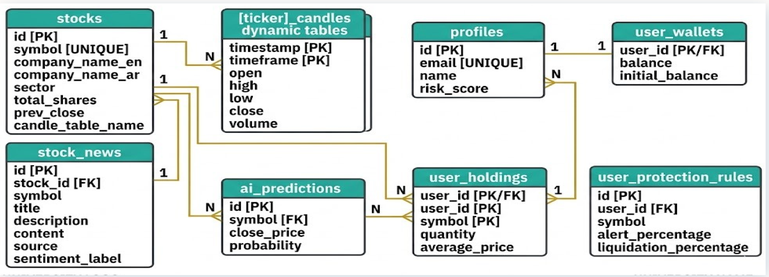



Figure 4.2: Relational Database Schema designed for high-scalability, featuring dynamic candle storage and isolated user portfolio management.

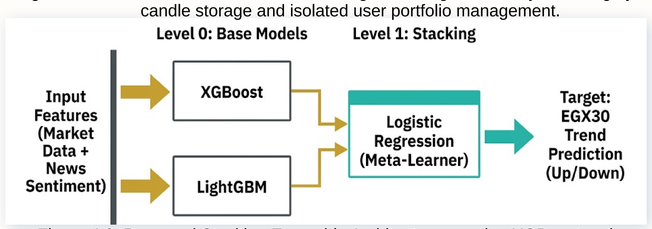



Figure 4.3: Proposed Stacking Ensemble Architecture merging XGBoost and LightGBM with a Logistic Regression Meta-Learner for optimized trend prediction.

## Lessons Learned

### Technical Expertise

- **Stacking**: Optimized XGBoost & LightGBM base learners.
- **Optuna**: Bayesian optimization for robust model stability.
- **Quant**: Hybrid technical and macroeconomic modeling.
- **Cloud**: Scalable deployment on AWS & Supabase.

### Core Knowledge

- **Target Engineering**: EMA10 noise filtering mechanics.
- **Quant Finance**: Gold, USD, and CBE rate correlations.
- **NLP**: Real-time Hugging Face sentiment pipelines.
- **Risk**: Designing automated stop-loss protection rules.

### Problem Solving

- **Bias**: Strict chronological splitting for temporal integrity.
- **Noise**: EMA-based smoothing for clean signal extraction.
- **Scale**: Aligning a massive 28-year historical dataset.
- **Latency**: AWS Lambda optimization for real-time inference

## Future Work

**Live Brokerage Integration**: Migrating from paper-trading to real-world execution via local brokerage APIs for automated trade execution.

**Multi-Index Expansion**: Adapting the Stacking Ensemble model to forecast broader indices like EGX70 and EGX100 for diversified stock

**SHAP-Based Explainability**: Utilizing Explainable AI (XAI) to show users exactly how macroeconomic factors influenced each signal.

## Conclusion

EGX360 achieves 90% accuracy by synergizing Egypt-specific macroeconomic indicators with a robust Stacking Ensemble. This project bridges the gap between raw data and actionable intelligence, providing retail investors with a production-ready tool for market volatility.